Note 5.4 The Fundamental Theorem of Calculus

Mean Value Theorem for Definite Integrals

- **Theorem 3-- The Mean Value Theorem for Definite Integrals:**
  - **Definition:** If f is continuous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\, dx$$

  - Proof: If we divide both sides of the Max-Min Inequality by $(b-a)$, we obtain

$$\min f \leq \frac{1}{b-a} \int_a^b f(x)\, dx \leq \max f$$

    Since f is continuous, then by intermediate value theorem, there must be some points c in $[a, b]$ satisfying $f(c) = \frac{1}{b-a} \int_a^b f(x)\, dx$.

Fundamental Theorem, Pt 1

- **Theorem 4-- The Fundamental Theorem of Calculus, Pt 1:**
  - **Definition:** If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)\, dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$:

$$F'(x) = \frac{d}{dx} \int_a^x f(t)\, dt = f(x)$$

    The theorem actually connect integrals and derivatives. If f is any continuous function, then then there exists a differentiable function F whose derivative is f.
  - Example: Use the Fundamental Theorem to find dy/dx if
    - $$y = \int_1^{x^2} \cos t\, dt$$
    - Sol: Let $u = x^2$, then

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= \left(\frac{d}{du} \int_1^u \cos t\, dt \right) \frac{du}{dx} \\ &= \cos u \cdot \frac{du}{dx} \\ &= 2x \cos x^2 \end{aligned}$$

- $$y = \int_{1+3x^2}^{4} \frac{1}{2+e^t}\,dt$$

- Sol:

$$\begin{aligned}\frac{d}{dx}\int_{1+3x^2}^{4} \frac{1}{2+e^t}\,dt &= \frac{d}{dx}\left(-\int_{4}^{1+3x^2} \frac{1}{2+e^t}\,dt\right) \\ &= -\frac{d}{du}\int_{4}^{1+3x^2} \frac{1}{2+e^t}\,dt \cdot \frac{du}{dx} \\ &= -\frac{1}{2+e^{1+3x^2}}\frac{d}{dx}(1+3x^2) \\ &= -\frac{6x}{2+e^{1+3x^2}}\end{aligned}$$

- Proof:
  Let $f(t)$ be an integrable function over a finite interval I, then fix an $a \in I$ and another number $x \in I$. Then we can define a new function F whose value at x is

$$F(x) = \int_a^x f(t)\,dt$$

The derivative of $F(x)$ is

$$\begin{aligned}F'(x) &= \lim_{h\to 0}\frac{F(x+h)-F(x)}{h} \\ &= \lim_{h\to 0}\frac{1}{h}\left[\int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt\right] \\ &= \lim_{h\to 0}\frac{1}{h}\int_x^{x+h} f(t)\,dt\end{aligned}$$

Then by the mean value theorem, there is a c in $[x, x+h]$ satisfying

$$f(c) = \frac{1}{h}\int_x^{x+h} f(t)\,dt$$

As $h \to 0$, $x+h \to x$ and $c \to x$. Since f is continuous at x,

$$\lim_{h\to 0} f(c) = f(x)$$

Thus

$$F'(x) = \lim_{h\to 0} f(c) = f(x)$$

Fundamental Theorem, Pt 2 (The Evaluation Theorem)

- **Theorem 4-- The Fundamental Theorem of Calculus, Pt 2:**
  - **Definition:** If f is continuous at every point in $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

  - Proof:
    By Pt.1 , we can suppose an antiderivative of f namely

$$G(x) = \int_a^x f(t)\,dt$$

    Since $F'(x) = f(x)$, then $F(x)$ is another antiderivative of f, thus $F(x) = G(x) + C$.
    Then

$$\begin{aligned} F(b) - F(a) &= [G(b) + C] - [G(a) + C] \\ &= G(b) - G(a) \\ &= \int_a^b f(t)\,dt - \int_a^a f(t)\,dt \\ &= \int_a^b f(t)\,dt - 0 \\ &= \int_a^b f(t)\,dt \end{aligned}$$

    since the variables in the integral is dummy variables, we can arbitrarily replace the variables.
  - Process:
    - Find an antiderivative F of f, and
    - Calculate the number $F(b) - F(a)$, which is equal to $\int_a^b f(x)\,dx$.
  - Notation:
    The notation for the difference $F(b) - F(a)$ is

$$F(x)]_a^b \quad \text{or} \quad [F(x)]_a^b$$

    depending on whether F has one or more terms.

The Integral of a Rate

- **Theorem 5-- The Net Change Theorem:**
  The net change in a function $F(x)$ on $[a, b]$ is the integral of its rate of change:

$$F(b) - F(a) = \int_a^b F'(x)\,dx$$

  (It can be obtain by take the $F'(x) = f(x)$ to the evaluation theorem.)

The Relationship between Integration and Differentiation

- The processes of integration and differentiation are "inverses" of each other. And it's important to find the antiderivative in order to evaluate definite integral easily.

Total Area

- **Strategy:** To find the area between the graph of $y = f(x)$ and the x-axis on $[a, b]$:
  - Subdivide $[a, b]$ at the zeros of f.
  - Integrate f over each subinterval.
  - Add the absolute values of the integrals.

Exercise

- P333 13 19 23 39 69 79 81
- Question 69th Initial Value Problems
  For the initial value problems

$$\frac{dy}{dx} = f(x), y(x_0) = y_0$$

  the solution can be expressed as

$$y(x) = \int f(x)\, dx + C$$

  also,

$$y(x) = \int_{x_0}^{x} f(t)\, dt + y_0$$

  since in the initial value problems, by the evaluation theorem,

$$\begin{aligned} \int_{x_0}^{x} f(t)\, dt &= F(x) - F(x_0) \\ &= \int f(x)\, dx + C_0 - F(x_0) \end{aligned}$$

  where $F(x)$ is an antiderivative of $f(x)$.
  then

$$y(x) = \int f(x)\, dx + C_0 - F(x_0) + y_0$$

which is equivalent to

$$y(x) = \int f(x)\,dx + C$$

where $C = C_0 - F(x_0) + y_0$.
So for this problem, express $y(x)$ in the definite integral form,

$$y(x) = \int_2^x \sec t\,dt + 3$$